

Software Analysis Report

Smart Bank Routing System

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1. Project Overview

1.1. Problem Statement

Customers often travel to the nearest Automated Teller Machine (ATM) expecting to perform a banking transaction, only to discover that the ATM is out of service, out of cash, or does not support the requested transaction type. This results in wasted time, customer frustration, and unnecessary congestion at bank branches.

The Smart Bank Routing System aims to solve this problem by directing customers to the nearest operational ATM or branch capable of fulfilling their requested service while providing real-time information regarding ATM availability, supported services, and cash status.

2. Objectives

The proposed system aims to achieve the following objectives:

- Reduce customer travel time.
- Eliminate unnecessary visits to unavailable ATMs.
- Provide real-time ATM operational status.
- Improve customer satisfaction.
- Assist the bank in monitoring ATM performance.
- Optimize ATM utilization across different regions.

3. Stakeholders

3.1. Customers

Customers use the application to locate suitable ATMs and receive navigation assistance.

3.2. Bank Administrators

Administrators manage ATM information, monitor availability, and update ATM records.

3.3. Maintenance Team

Maintenance personnel update ATM operational status after maintenance activities and report technical issues.

4. Functional Requirements

The following functional requirements describe what the system must accomplish.

4.1. Customer Functions

1. Register and login.
2. View nearby ATMs.
3. View ATM operational status.
4. View supported transaction types.
5. Obtain navigation to the selected ATM.
6. Report incorrect ATM status

4.2. Administrator Functions

1. Add / Update / Remove ATMs.
2. Update ATM status.
3. Update supported services.
4. Update cash availability.

4.3. Maintenance Functions

1. Mark ATM as under maintenance.
2. Mark ATM as operational.
3. Report hardware failures.
4. Record maintenance history.

5. Non-Functional Requirements

Performance	Nearby ATM search should complete within two seconds under normal operating conditions.
Availability	The system should be available 24 hours a day, seven days a week.
Scalability	The system should support thousands of ATMs distributed across multiple cities.
Reliability	ATM status updates should accurately reflect real-world conditions with minimal delay.
Security	Customer information and location data must be securely stored and transmitted using encryption.
Usability	The application should be simple enough for users of all ages.
Maintainability	System modules should be designed independently to simplify future enhancements.
Compatibility	The system should support Android, iOS, and modern web browsers.

6. System Actors

Actor	Description
Customer	Searches for nearby ATMs and branches.
Administrator	Manages ATM information.
Maintenance Staff	Updates ATM operational status.
Mapping Service	Provides routing information.
ATM Monitoring System	Supplies real-time ATM status.

7. Major Use Cases

1. Find Nearby ATM
2. Search Nearby Branch
3. Get Best Route
4. View ATM Details
5. Update ATM Status
6. Update Cash Availability
7. Report ATM Issue

8. Preliminary Database Entities

The following entities are expected to exist during database design.

Entity	Purpose
Customer	Registered users
ATM	ATM information
Branch	Bank branch information
ATMStatus	Operational status
TransactionType	Supported banking services
CashInventory	Remaining cash inside ATM
MaintenanceRecord	Maintenance history

9. Business Rules

The system follows the following business rules.

1. Only operational ATMs may be recommended.
2. The ATM must support the requested transaction type.
3. Cash withdrawal recommendations require sufficient available cash.
4. Offline ATMs must not appear in routing results.
5. ATM status updates require administrator or maintenance authorization.
6. Customer location must be obtained with permission.

10. Possible External APIs

The proposed solution may integrate with the following services.

- Google Maps API

11. Proposed System Architecture

The system consists of the following components:

1. Web Application
2. ASP.NET Core REST API
3. SQL Server Database
4. ATM Monitoring Service
5. Routing Service
6. Administration Dashboard

12. Assumptions

The analysis assumes the following:

- Every ATM periodically reports its operational status.
- Cash inventory is automatically synchronized with the central banking system.
- GPS services are available on customer devices.
- Customers grant location permission.
- The banking network securely communicates with every ATM.

13. Conclusion

The Smart Bank Routing System addresses a common banking inconvenience by enabling customers to quickly locate the nearest operational ATM or branch that supports their desired banking service. Through real-time ATM monitoring, intelligent routing, and administrative management tools, the system improves customer satisfaction while reducing unnecessary travel and enhancing operational efficiency. The proposed analysis establishes a solid foundation for the subsequent design, implementation, and testing phases of the software development lifecycle.